MIL-R-24037A
12 December 1969
SUPERSEDING
MIL-R-24037(SHIPS)
29 May 1964 and
MIL-R-10059D
25 January 1954

MILITARY SPECIFICATION

RADIACMETERS, QUARTZ-FIBER SELF-INDICATING GENERAL SPECIFICATION

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1 1 This specification covers ion-chamber quartz-fiber dosimeters (radiacmeters) capable of measuring doses of gamma or X-radiation

2 APPLICABLE DOCUMENTS

2 1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566	—	Boxes, Folding, Paperboard
PPP-B-636	—	Box, Fiberboard
PPP-B-676	—	Boxes, Setup
PPP-T-76	—	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, for Carton Sealing
PPP-T-97	—	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	—	Preservation, Methods of
MIL-A-8625	—	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-P-15024	—	Plates, Identification - Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment
MIL-E-21981	—	Electronics Type Designation, Identification Plates and Markings, Requirements for
MIL-C-23237	—	Charger, Radiac Detector
MIL-I-45208	—	Inspection System Requirements
MIL-C-55038	—	Charger, Radiac Detector PP-1578( )/PD

STANDARDS

MILITARY

MIL-STD-105	—	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	—	Marking for Shipment and Storage
MIL-STD-167	—	Mechanical Vibrations of Shipboard Equipment
MIL-STD-202	—	Test Methods for Electronic and Electrical Component Parts
MIL-STD-454	—	Standard General Requirements for Electronic Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6665

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2.2 Other publications - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, Room 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3 REQUIREMENTS

3.1 Qualification. - Quartz fiber dosimeters furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list (see 6.2)

3.2 Detail requirements for individual quartz-fiber dosimeters. - Detail requirements or exceptions applicable to individual types of quartz-fiber dosimeters shall be as specified in the detail specifications listed in Supplement 1 to this specification. In the event of any conflict between requirements of this specification and the detail specification, the latter shall govern

3 3 Requirements to be included in detail specifications - The following requirements shall be as specified in the detail specifications·

(a) Scale marking.
(b) Range
(c) Leakage and leakage after heavy irradiation
(d) Accuracy.
(e) Energy dependence.
(f) Ion chamber saturation and rate dependence.

3.4 Reliability - The equipment will be subjected to continuous use for long periods under the varied and severe conditions of Military service without overhaul If it is to meet these conditions, it is imperative that reliability of operation be considered of prime importance in the design and manufacture of the equipment. The manufacturer shall employ all methods possible in the process of manufacture which will assure quality and maximum reliability parts to equipment circuits, adequate factors of safety shall be provided by suitable deratings from the part specification values in order to insure high equipment reliability under all service conditions The design shall include all possible features which will result in reliable and stable operation.

3 5 Materials - Materials shall be chosen with care, not only from a standpoint of complete suitability for the purpose intended, but also based on the following three considerations·

(a) Minimal response to neutrons
(b) Minimum use of materials with appreciable neutron cross-sections wherever possible.
(c) Proper energy dependence as specified in the individual detail specification.

3.6 Description. - The radiacmeter shall be a hermetically-sealed, pocket-type quartz-fiber electroscope for detecting and indicating integrated exposures to X- and gamma radiations The electroscope shall discharge by an amount proportional to the quantity of radiation which passes through the chamber. The total amount of radiation to which the instrument has been exposed shall be indicated by the image of the electroscope fiber on a calibrated reticle It shall be possible to observe the position of the fiber image on the reticle when the radiacmeter is pointed at a source of light. The instrument shall have a cylindrical barrel equipped with an alligator type pocket clip.

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3.7 Mechanical. -

3.7.1 Electroscope fiber. - The electroscope fiber shall be constructed of quartz or any equivalent material approved by the command or agency concerned

3.7.2 Scale - The reticle shall be etched or engraved, or applied by the glue silver process or any other process approved by the command or agency concerned on a diaphragm of glass or other material satisfactory to the command or agency concerned The scale shall be unaffected by strong sunlight or the range or storage temperatures specified in 3 10.

3.7.2 1 Imperfections or spots - Imperfections or spots, on the scale (a) shall not be wider than the width of a major scale line in the rectangular area defined by the scale graduations, (b) spots external to this area may be of such size and number so that the integrated total area of these spots shall appear no greater than the area of the zero mark on the scale. For this purpose only those spots having a diameter larger than the width of a major division line shall be counted.

3 7.2.2 Scale division spacing. - Using the zero and central readings as a datum, scale division spacings shall not deviate more than 5 percent at the 1/4 and 3/4 scale levels and not more than 10 percent at the full scale level

3.7 3 Charging electrode diaphragm - The charging electrode diaphragm shall be constructed of a flexible, high ohmic resistant material. The charging electrode shall be mounted in the center of the diaphragm The diaphragm shall be so constructed that the force required, at sea level and at an ambient temperature of 70° ±5°F., on the charging electrode to make electrical contact between it and the internal electrode leading to the electroscope shall be greater than 2 pounds, but shall not exceed 6 pounds

3 7.3.1 External portion of the charging electrode. - The length of the external portion of the charging electrode shall be minimized, and the electrode shall be capable of withstanding axial forces of at least 10 pounds and radial forces of at least 7 pounds. The diaphragm shall be recessed in the end of the barrel to avoid accidental discharge This recess between the end of the barrel and the diaphragm shall have the following dimensions

Diameter - 0.365 ±0 005 inch.
Depth (to charging electrode whenever in contact with the internal electrode, that is, when the electrode is loaded with a force of not more than 6 pounds) — 0.220 ±0.030 inch.

3 7 3 2 Charging socket resistance - The resistance of the diaphragm shall be at least $1x10^{12}$ ohms

3.7.3.3 Charging contact resistance. - The contact resistance of the electroscope switch, the charging electrode, the charging end of the dosimeter, and the inside surface of the charging recess shall be as low as possible. (For tests, a fully charged dosimeter shall discharge to full scale reading in less than one-half second.)

3.7.4 Radiacmeter barrel. - The barrel shall be metal and shall have dimensions not greater than 4-1/2 inches in length and 9/16 inch in diameter

3.7 4.1 External surface. - The external surface of the radiacmeter shall be smooth, with no grooves or crevices, with the exception of the groove for the clip specified in 3.7.6. The bottom rim of the barrel shall make electrical ground contact with the ridge in the charger receptacle when used with Radiac Detector Charger PP-1578( )/PD and PP-4726( )/PD specified in MIL-C-55038 and MIL-C-23237 respectively

3 7.5 Finish - If the barrel is made of aluminum, the finish shall be black anodize in accordance with type II, Class 2 of MIL-A-8625 or as specified in the contract or order (see 6.1 1). For any other barrel material, finish shall be equivalent in smoothness, porousness, and reflectivity and shall be specifically approved by the command or agency concerned

3 7 6 Clip - Unless otherwise specified (see 6 1 1) the radiacmeters shall be provided with a double jaw alligator type metal clip with a coil spring spot welded (two points) to a collar no less than 1/4 inch in width.

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The clip shall be removable but so mounted it cannot be easily slipped off or rotated about the barrel When clipped to a cotton shirt pocket, the clip shall withstand a force of 4 pounds without sufficient slippage to disengage the clip from the cloth This requirement shall be met after opening and closing the jaws 1000 times. The color of the clip shall be the same as the barrel.

3.7.7 Government nomenclature - Nomenclature shall be requested in accordance with the procedures outlined in MIL-E-21981

3.7.7 1 Marking information - Marking information shall be in accordance with MIL-P-15024 except the barrel shall be marked as follows.

IM-____( )/PD Radiacmeter
Contract or Order No.____.
Manufacturer's designating letters ____.
Serial No ____

3 7.8 Weight - The complete radiacmeter shall not exceed 2 ounces in weight

3.8 Electrical and performance -

3.8 1 Electroscope charging voltage.- The equipment shall be charged by an auxiliary electrostatic or battery-operated charger, of the types specified in 3.7.4.1 The total charging voltage of the electroscope and its system shall not exceed 250 volts and not less than 145 volts. The dosimeter shall be capable of withstanding a maximum of 200 volts ±5 percent of positive charge without permanently distorting, kinking, bending, or electrically rupturing or breaking down the fiber

3 8 2 Optical system - The optical system shall allow the fiber image and reticule scale to be clearly resolved by the normal, dark-adapted eye, without parallax, when exposed to an axial transmitted illumination of 0 5 foot candle The focus of fiber and reticule shall permit resolution to 1/4 of the smaller scale division over the entire scale The image of the fiber shall appear as a vertical line to within 1/4 of a minor scale division over the entire scale, that is, with the charging potentials adjusted so that the image of the fiber coincides with the bottom of any major scale division marking, the distance between the image of the fiber and the top of that same major division marking shall not be greater than 1/4 of the distance between two adjacent minor division markings

3.8 2 1 Material for scale - The scale shall be constructed of a material which will not become discolored, pitted, or damaged in any way by exposure to light, heat, or any ionizing radiation

3.8.3 Response to beta radiation.- Unless otherwise specified in the contract or order (see 6 1.1), response to beta radiation in absence of gamma radiation shall be not greater than 15 percent (that is, a dose in rads beta radiation shall not cause a reading of greater than 0.15 times this dose in roentgens on the radiacmeter) for any beta energy up to the energy of strontium-yttrium betas This shall be accomplished by proper wall thickness surrounding the ion chamber.

3.8 4 Life - The radiacmeter shall be capable of operating as specified herein through 2,500 charging operations when tested as specified in 4.6 15.

3.8.5 Geotropism - The deflection from the normal (fiber image in the vertical position) reading of the electroscope fiber image during a 360-degree rotation of the radiacmeter about its longitudinal axis, while that axis is perpendicular to the force of gravity shall not exceed plus or minus one scale division

3.8.6 Change in reading after charging operation - The radiacmeter shall be so constructed that the total change in reading due to (a) breaking contact between the diaphragm and electroscope electrodes, (b) removal of radiacmeter from charging receptacle, and (c) grounding of the diaphragm electrode to barrel after removal from the charger, shall not exceed one scale division

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3.9 Service and environmental operating conditions - All requirements of 3 8 shall be met under any of the following environmental conditions, either separately or in any combination, for at least the length of time specified for each inspection of 4.5:

(a) Temperature - A temperature range of minus 40° ±5° to plus 131° ±5°F.
(b) Relative humidity - Any relative humidity up to 100 percent
(c) Elevation - Any pressure between atmospheric and a pressure equivalent to 50,000 feet above sea level
(d) Immersion - Any depth of water up to 3 feet
(e) Salt spray - Under conditions of salt spray as specified in 4 6 21
(f) Rough handling - When worn on the clothing, during any activity of the wearer, however strenuous Under the shock test specified in 4 6 10 a reading shall not change by more than one scale division.
(g) Thermo mechanical equilibrium - Within the temperature range of -40°F to +131°F, shifts of the fiber, relative to the scale resulting from relative contractions or expansions of dosimeter components due to the temperature variation, shall not cause an apparent deflection exceeding ±5 percent of full scale deflection at +131°F and ±8 percent of full scale deflection at -40°F

3 10 Environmental requirements, nonoperating - The radiacmeter shall be capable of fulfilling the requirements of 3 8 after exposure to the following

(a) Temperature - Storage at minus 80° ±5°F
(b) Vibration - After the virbation test specified in 4 6.22.

3 11 Workmanship - Workmanship shall be in accordance with Requirement 9 of MIL-STD-454

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4 1.1 Contractor's inspection system The contractor shall provide and maintain an effective inspection system acceptable to the Government covering the supplies under the contract. The inspection system shall be in accordance with specification MIL-I-45208. A current written description of the system shall be submitted to the cognizant Government inspector for evaluation of acceptability (see 6.1.2) The written description shall include flow charts showing the complete flow of material from inspection on receipt, through all manufacturing processes, to final shipment. The various processes and inspection positions shall be clearly identified on the flow charts and cross referenced to the detailed inspection procedures and criteria of the written description of the inspection system. Any changes to the approved inspection system which might affect the degree of assurance required by this specification or other applicable documents shall be submitted to the cognizant Government inspector for approval prior to use.

4 1.2 Government verification All quality assurance operations performed by the contractor will be subject to Government verification at any time Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with this or the individual specification The amount of verification will be adjusted to make maximum utilization of the contractor's quality assurance operations and the quality history of the product Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated

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4.2 Classification of inspections.- The inspection requirements specified herein are classified as follows

(a) Qualification inspection (see 4.3 1)
(b) Quality conformance inspection (see 4.5)

(1) Production inspection
(2) Aged product inspection

(c) Retention of qualification inspection (see 4 4)
(d) Inspection of preparation for delivery (see 4 5 4)

4.3 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Electronic Systems Command. Qualification tests shall consist of the tests specified in table I, and the detail specification. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.2 and 6.3)

TABLE I - Qualification inspection

Item	Reference		Submission of data required
	Requirement	Test	
Design and construction	3 6, 3.7 3 3 (range), 3 8.1, 3 8.2, 3 8.5, 3.8 6 and 3 11	4.6 1 4.6.2 4 6 3 4 6 8 4 6 13	X
Leakage	3 3	4 6.4	X
Accuracy	3 3	4.6.6	X
Immersion	3 9	4.6.9	X
Shock, rough handling	3 9	4.6.10	X
Energy dependence	3 3	4.6.11	X
Leakage after heavy irradiation	3.3	4.6.12	
Response to beta radiation	3.8.3	4 6 14	
Life	3.8.4	4.6 15	X
Ion chamber saturation and rate dependence	3.3	4.6.16	
Temperature, operating	3 9	4 6 17	X
Temperature, ono-operating	3 10	4 6 18	X
Humidity	3.9	4.6.19	X
Salt spray	3.9	4.6.20	
Elevation	3 9	4 6 21	X
Vibration	3.10	4.6.22	
Thermo mechanical equilibrium	3 9	4 6 23	
Charging socket resistance	3.7 3.2	4.6.24	
Charging contact resistance	3 7.3.3	4.6.25	

4 3 1 Qualification inspection data - Qualification inspection data to indicate performance in those areas shown in table I shall be submitted to the command or agency concerned.

4.3.2 Sample submission procedures - This details the procedure for submission of samples, with related data, for qualification inspection of quartz fiber, self-indicating dosimeters covered by this specification The procedure for extending qualification of the required sample to other dosimeters covered by this specification is also outlined herein.

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4.3.2.1 Samples - Quartz-fiber dosimeters which fall in the range of 0 through 1.0 R are considered low-range dosimeters Quartz-fiber dosimeters which cover the ranges above 1.0 roentgen (R) are considered high-range dosimeters

4 3 2 2 Single-type submission - A sample consisting of 20 sample units of any dosimeter which falls within the range of a low-range or a high-range dosimeter shall be considered a single type submission

4 3 2 3 Combined submission - When qualification of both the low range and high range is desired, 20 sample units of a low range and 20 sample units of a high range shall be submitted.

4 3.2.4 Test data - When tests are to be performed at a Government laboratory, prior to submission all sample units shall be subjected to all the tests specified in table I Each submission shall be accompanied by the test data obtained from these tests All tests shall be submitted in duplicate.

4.4 Retention of qualification To retain qualification, the supplier shall forward a report at 24 month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

a. A summary of the results of the tests performed for inspection of product for delivery, group A and B, indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for

b A summary of the results of tests performed for qualification verfication inspection, group C, including the number and mode of any subgroup failures. The summary shall include results of all qualification verification inspection tests performed on completed lots during the 24 month period. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

4.4 1 Failure to submit the report within 90 days after the end of each 24 month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the supplier shall immediately notify the qualifying activity at any time that the inspection data indicates noncompliance of the product to meet the requirements of this specification.

4.4.2 In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during 2 consecutive reporting periods there has been no production, the manufacturer may be rqeuired, at the discretion of the qualifying activity, to submit the radiacmeters to testing in accordance with the qualification inspection requirements

4 5 Quality conformance inspection -

4.5.1 Production inspection.- This inspection shall consist of group A, B, and C inspections Sampling shall be as specified herein

4 5 1 1 Sampling for lot inspections.-

4 5 1.1.1 Inspection lot - One month's production shall be considered a lot for purposes of inspection and testing.

4.5.1.1.2 Sampling for group A inspection.- A random sample of radiacmeters shall be selected from each inspection lot and shall be subjected to each of the group A tests specified in 4.5.1.2 with lot acceptance based on the sampling requirements shown in table III in accordance with MIL-STD-105.

4.5.1.1.3 Sampling for group B inspection.- A random sample of radiacmeters shall be selected from each inspection lot in accordance with table IV, and shall be subjected to each of the group B tests specified in 4.5.1.3.

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4.5.1.1.4 Sampling for group C inspection.- A sample number of radiacmeters in accordance with table II shall be selected from the end of each month's production and subjected to the group C inspection specified in 4 5.1 4. Performance of group C tests are the responsibility of the contractor. They may be performed and certified by an approved testing laboratory.

TABLE II Sampling for group C inspection

Number of radiacmeters in 1 month's production	Number of radiacmeters for group C inspection
500 and under	3
501 to 1,300	5
1,301 to 8,000	7
8,001 and over	10

4.5.1.2 Group A inspection.- These inspections (including sampling) shall be for the requirements shown in table III and the General Inspection Level II of MIL-STD-105. Group A inspection shall be performed in any order except that mechanical visual inspections shall be performed before the performance inspections and the immersion test (see 4.6.9) shall be conducted last. Separate random samples shall be selected for each of the subgroups. In the event of a lot failure, 100 percent of the units in the lot (except for the acceptable samples) shall be screened for all requirements listed under that specific sub-group

Table III Group A inspection

Inspection	Requirement Paragraph	Inspection Paragraph	AQL	
			Major	Minor
Visual and mechanical	3 3 (range)			
Visual	3 6	4 6 1	1 0%	4.0%
	3.7 1	4.6.1		
	3.7.2	4 6.1		
	3 7 3	4 6.1		
	3.7 5	4.6.1		
	3.7 6	4 6 1		
	3 7.7	4 6 1		
	3 7 8	4.6 1		
	3.11	4.6.1		
Mechanical			1 0%	
Dimensions				
Barrel dimensions	3 7 4	4 6.2		
		4 6.3		
Effect of axial force on charging electrode	3 7 3 1	4 6 8		
Sub-group I				
Charging electrode diaphragm	3 7 3	4 6 2	1.0%	
Sub-group II				
Charging voltage	3.8 1	4 6.2		
Charging socket resistance	3 7 3 2	4 6 24		
Change in reading after charging	3.8.6	4 6 2	1 0% for the group	
Geotropism	3 8 5	4 6 7		
Fiber optics	3.8 2	4 6.13		
Calibration - midscale	3.3	4 6 5		
Charging contact resistance	3 7.3 3	4 6.25		

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Table III. Group A inspection (continued)

Inspection	Requirement Paragraph	Inspection Paragraph	AQL	
			Major	Minor
Sub-group III				
Electrical leakage	3 3	4 6.4	1 5%	
Sub-group IV				
Immersion	3.9	4.6 9	1.0%	

4.5 1.3 Group B inspection - These inspections (including sampling) shall be for the requirements shown in the Table IV. The inspection level shall be S3 of MIL-STD-105. Group B inspections shall be performed on inspection lots that have passed group A inspections and specifically on samples which have been subjected to and which do meet the group A requirements Separate random samples shall be selected for each of the sub-groups. The tests shall be conducted in the order listed. In the event of a lot failure, 100 percent of tne units in the lot (excluding the acceptable samples) shall be screened for all requirements listed under that specific sub-group

Table IV Group B inspection

Inspection	Requirement Paragraph	Inspection Paragraph	AQL	
			Major	Minor
Sub-group I				
Shock	3 9	4.6.10	2 5%	
Sub-group II				
Thermo Mech Equilibrium	3 9	4 6.23	2 5%	
Sub-group III				
Elevation	3 9	4.6.21	2.5%	
Sub-group IV				
Leakage after heavy radiation	3 3	4.6.4	2.5%	
Calibration - entire scale	3.3	4 6 6	for the	
Precision	(see 3.3)		group	

4.5.1 4 Group C inspection.- These inspections (including sampling) shall be for the requirements shown in table II and table V Group C inspection shall be performed on inspection lots that have passed both group A and B inspections. Separate random samples shall be selected for each of the sub-groups Failure to conform to the specification requirements for any group C test shall cause the contractor to correct the cause of failure on future production units and to repair the deficiency in any radiacmeter not yet shipped. If investigation indicates that the defects may exist in items previously shipped, full particulars concerning the defect(s) found, including recommendations for correction, shall be furnished to the command or agency concerned

Table V Group C inspection

Inspection	Requirement Paragraph	Inspection Paragraph
Sub-group I		
Energy dependence	3 3	4 6.11
Operating temperature	3 9	4.6 17
Life	3 8.4	4.6.15
Sub-group II		
Salt spray	3 9	4 6.20
Humidity	3.9	4.6 19
Vibration	3 10	4 6.22
Rate dependence	3.3	4 6.16
Temperature nonoperating	3 10	4 6 18

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4.5 2 Aged product inspection. - Prior to delivery and after production inspection each radiacmeter shall be stored for a period of 30 days and then tested for conformance to leakage and accuracy requirements of 3 3 in accordance with 4 6 4 and 4 6.6 One copy of the results shall be enclosed with each lot shipped and one copy to the command or agency concerned

4 6 Test procedures -

4.6.1 Visual and dimensional examination. - A visual examination shall be performed to determine conformance with 3.6, 3 7 1, 3 7.2, 3.7.5, 3.7.6, 3.7.7, 3 7.8, (range) 3 3, 3 11 and the general construction of the diaphragm as specified in 3 7.3

4 6.2 Charging. - Using an appropriate jig, the radiacmeter shall be charged. The jig shall be so constructed that

(a) A pressure of greater than 6 pounds exerted by the radiacmeter charging pin will short the charger.

(b) A barrel of too large an outside dimension cannot enter the charging receptacle

(c) A radiacmeter charging-recess where the pin is at too great a depth will not make contact.

The charger arrangement used shall be incapable of delivering more than 190 volts. The radiacmeter shall be capable of being charged smoothly and with normal effort in this arrangement This jig and charging arrangement shall be approved by the command or agency concerned prior to commencement of production. In addition, the radiacmeter shall be tested for ease of charging with a PP-1578( )/PD and PP-4276( )/PD charger Upon removal from this charger, the change in reading shall be not greater than one scale division

4 6.3 Dimensions - The test specified in 4 6.2 will serve to show that

(a) The barrel diameter is not greater than the maximum

(b) The charging pin is not recessed too far within the barrel of the radiacmeter

An appropriate gauge shall be used to determine whether the barrel diameter and charging pin recess depth are within the opposite limit as well

4 6 4 Leakage, electrical - The test for electrical leakage shall be in accordance with the detail specification

4 6.5 Calibration at midpoint only. - The test for calibration at midpoint shall be in accordance with the detail specification.

4.6.6 Calibration over the entire range - A test shall be made to determine that the calibration over the entire range is in accordance with the detail specification

4 6 7 Geotropism - Geotropism shall be checked with the fiber at the midscale position. The radiacmeter shall be held horizontally and rotated about its axis The fiber position shall not change from the above values by greater than one scale division

4 6 8 Effect of forces on charging electrode. - By use of a suitable jig, the effect of a 10-pound axial force and a 7-pound radial force shall be determined The radial force shall be applied at right angles to the direction of motion of the diaphragm The applicable forces shall be applied ten times in each direction Failures shall be indicated by breaking or tearing of the diaphragm or bending of the pin

4 6 9 Immersion. - The radiacmeter fully charged at room temperature, shall be immersed for 2 hours at a depth of 3 feet in water maintained at the temperature of melting ice. There shall be no sign of leakage of water into the radiacmeter, and the reading shall not change by greater than ± 3 percent

4 6 10 Shock and rough handling - When fully charged, a radiacmeter shall be subjected to six drops from a height of 4 feet onto a hardwood floor, two on each end and two on the side At the end of this series, the reading shall not have changed in either direction by an amount greater than that corresponding to one scale division After each individual drop the reading shall change not more than one scale division

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4 6.11 Energy dependence. - Performance at midscale shall be checked at approximately 30, 50, 80, 120, 160, and 200 kiloelectrovolt (kV) Performance shall also be determined for each radiacmeter with cobalt 60 or cesium 137, for comparison purposes Tube voltages and filtrations for the particular energies shall be as specified by the National Bureau of Standards for a 250-KVP (kilovolt, peak) machine. Performance shall be in accordance with 3.3

4 6.12 Leakage after heavy irradiation - The radiacmeter will be tested to determine conformance with 3.3 (see 6.1.1)

4.6.13 Illumination of optical system. - Using an appropriate light source, performance shall be tested to determine compliance with 3.8.2 Tests shall take place in a dark room The fiber and scale shall be easily visible

4.6.14 Response to beta radiation (see 6.1.1). - A beta emitting isotope with beta energy of approximately 2.2 megaelectrovolt (MeV), shall be used. Strontium-yttrium 90 is acceptable. The indicated dose for a computed exposure beta dose of one half full scale shall be no greater than 15 percent

4.6.15 Life test - An appropriate jig and counter arrangement shall be constructed to permit flexing the diaphragm of each unit 2500 times. The radiacmeter shall perform in accordance with 4.6.2, 4.6.4, and 4 6 6 after this test. The life test shall be performed as the last test of group C, subgroup 1

4 6.16 Saturation and rate dependence - The radiacmeter will be subjected to x-rays or gamma rays for short, equal intervals of time at rates specified (see 3.3)

4.6.17 Temperature. - Sample instruments shall be subjected to the following temperature test It will be considered that stabilization of the instrument's temperature has been reached when three consecutive temperature readings, at 10-minute intervals, taken at suitable points of the instrument, show no more than a plus or minus 5°F. differential During the temperature cycle, the temperature shall be raised or lowered in steps of approximately 35°F Stabilization shall be effected at each step. The instrument shall comply with 3.3 (accuracy) at midscale only, and with 3.3 (leakage) throughout all of the following steps·

(a) Start at room temperature (70° to 90°F.), take necessary measurements to establish a standard of performance

(b) Raise the temperature to 131° ±5°F. and repeat the measurements taken in (a)

(c) Reduce to room temperature, repeat measurements taken in (a).

(d) Reduce to a minus 40° ±5°F., repeat measurements taken in (a)

(e) Raise to room temperature and repeat measurements taken in (a)

4.6.18 Temperature, storage.- Sample instruments shall be subjected to the following temperature storage test Test leakage, and accuracy of radiacmeter at 30, 60, and 80 percent of full scale. Reduce temperature to minus 80° ±5°F. Hold for 72 hours Return to room temperature Repeat above measurements Raise temperature to 160°F Hold for 72 hours Return to room temperature Repeat measurements All measurements shall be in accordance with 3.3 for accuracy. Upon completion of storage, the meter shall be subjected to a leakage test to determine conformance with 3.3 (leakage)

4.6.19 Humidity test.- Sample instruments shall be subjected to the following humidity test·

(a) Start at room temperature 70° to 90°F and a relative humidity of 50° ±5 percent, take necessary measurements to establish a standard of performance.

(b) Raise the temperature to 131° ±5°F and a relative humidity to a minimum of 95 percent and hold for 24 hours. Reduce to room temperature and a relative humidity of 50 percent ±5 percent and repeat measurements taken in (a).

(c) Repeat (b) and test in accordance with 3 3 (leakage)

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4.6.20 Salt spray.- Sample instruments shall be given a 96-hour salt spray test in accordance with method 101 of MIL-STD-202 There shall be no evidence of corrosion following this test Performance will be determined before and after, but not during the test.

4 6 21 Elevation - Instruments shall be tested under reduced pressure at ambient temperature. An absolute pressure equivalent of less than 0.1 millimeter of mercury shall be maintained for a minimum of 72 hours during this test The instruments shall be required to pass the tests specified in 4.6.5 while still under reduced pressure after 54 hours, and pass the tests specified in 4.6.4 and 4 6.6 after return to atmospheric pressure.

4 6 22 Vibration - The type I vibration tests of MIL-STD-167 shall be performed

4 6.23 Thermo mechanical equilibrium tests - The instrument shall be zeroed at room temperature and then placed at 131°F for sufficient time to insure a condition of thermal equilibrium (approximately 2 hours) at which time they are read, and then returned to room temperature for 2 hours and read again. This test shall be performed in an identical manner at -40°F. The requirement of 3.9(g) shall be met

4.6.24 Charging socket resistance.- The resistance of the diaphragm, as measured with a Kiethly type 610A Electrometer or equal between the central charging pin and the barrel shall be at least (no less than) 1×10^{12} ohms with zero force applied to the charging diaphragm (with charging contact open).

4.6 25 Test for compatibility with radiacmeter charger.- The radiacmeter shall be depressed on the charging pedestal of Radiac Charger PP-1578A or PP-4276 ( )/PD and charged to zero Units shall comply with requirements of 3.8.6

4 7 Preservation and packaging inspection - Inspection of level A packages shall be in accordance with MIL-P-116

4 7 1 First article inspection - First article inspection, including rough handling test, is required except where level C packing and packaging is specified First article inspection will not be required when satisfactory evidence is furnished to the Government that items have been prepared for shipment identically with a previously approved package and pack for the same or similar item

4 7.2 Inspection of preparation for delivery.- The inspection of the preservation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5.

5 PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be Level A or C, as specified (sec 6.1.1)

5 1 1 Level A -

5.1 1 1 Cleaning.- Radiacmeters shall be cleaned in accordance with process C-1 of MIL-P-116

5 1 1 2 Drying - Radiacmeters shall be dried in accordance with the applicable procedure of MIL-P-116.

5 1 1 3 Preservation application - None required

5.1 1 4 Unit packaging.- Each radiacmeter shall be packaged in accordance with Method III of MIL-P-116 Fold one copy of the technical instructions in half across the long dimension and wrap securely around the radiacmeter Place the Radiacmeter, with the technical instructions wrapped around it, within a close fitting box conforming to PPP-B-566, variety 2, or PPP-B-676, variety 2 Close the box in accordance with the appendix of the applicable box specification

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5.1.1.5 Intermediate package - Fifty radiacmeters, packaged as specified in 5.1 1 4, shall be arranged in a most compact manner within a close fitting box conforming to PPP-B-636, W5c Closure shall be in accordance with the appendix of the box specification In addition, seal all seams and joints with tape conforming to PPP-T-76. No intermediate packaging is required when the total quantity shipped to a single destination is less than 50 units.

5.1.2 Level C.- Each radiacmeter shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity

5.2 Packing.- Packing shall be Level A, B, or C, as specified (see 6.1.1).

5.2.1 Level A - A quantity of radiacmeters packaged as specified in 5.1, shall be packed in a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant, style optional, special requirements use Closure shall be in accordance with the appendix to the box specification The shipping containers shall be reinforced by filament tape banding conforming to PPP-T-97, type IV.

5.2.2 Level B.- A quantity of radiacmeters, packaged as specified in 5.1, shall be packed in a close-fitting fiberboard box conforming to PPP-B-636, type CF, class domestic. Closure shall be in accordance with the appendix of the applicable box specification

5.2.3 Level C.- Radiacmeters, packaged as specified in 5.1, shall be packed in shipping containers in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129

6 NOTES

6 1 Ordering data - Procurement documents should specify the following

6.1.1 Procurement requirements.-

(a) Title, number and date of this specification

(b) Title, number and date of the applicable detail specification

(c) Finish, if other than black anodize (see 3.7.5)

(d) Chip requirements (see 3.7.6).

(e) Beta response (see 3 8 3).

(f) That the leakage after heavy irradiation (see 4 6.12), response to beta radiation (see 4 6.14), saturation and rate dependence (see 4.6 16), test will be performed by the Government except leakage after heavy radiation for the IM-9( )/PD

(g) Selection of applicable levels of packaging and packing required (see 5 1 and 5.2)

6 1.2 Contract data requirements Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423). The data required by this specification includes, but is not restricted to the following

(a) Contractor's inspection system (see 4 1.1)

(b) Test data (see 4.3.2 4)

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6.2 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List 24037, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification The activity responsible for the Qualified Products List is the Naval Electronics Systems Command, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be obtained from that activity Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6 3)

6.3 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

Custodians
Army - EL
Navy - EC
Air Force -

Review activities
Army -
Navy - SH, MC, CG
Air Force -

User activities
Army -
Navy -
Air Force -

Preparing activity
Navy - EC

(Project 6665-0249)

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NAVAL ELECTRONIC SYSTEMS COMMAND
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2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

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